

Rates of Change continued

44. The liquid in a vessel flows out through a tap at a rate given by $\frac{dv}{dt} = -10(40-t)$ where $v \text{ cm}^3$ is the volume of liquid in the vessel at time t seconds. Initially the vessel contained 8000 cm^3 of liquid. Find the volume of liquid in the vessel after 10 seconds.

$$\begin{aligned} \frac{dv}{dt} &= -10(40-t) \\ \frac{dv}{dt} &= -400 + 10t \\ \text{Integrate on both sides} \\ v &= -400t + 10 \frac{t^2}{2} + C \\ v &= -400t + 5t^2 + C \\ v(0) &= 8000 \\ 8000 &= -400(0) + 5(0)^2 + C \\ C &= 8000 \end{aligned}$$

$$\begin{aligned} V(t) &= -400t + 5t^2 + 8000 \\ t=10, V(10) &= -400(10) + 5(10)^2 + 8000 \\ &= 4500 \text{ cm}^3 \end{aligned}$$

45. The flow rate of water into a dam over the course of a particular year varies with time and is modelled by $\frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$, where W is the volume of water in the dam measured in thousands of cubic metres and t is the time in months measured from the beginning of January.

- (a) Find the maximum flow rate and the time at which the maximum flow rate occurs.
(b) Given that the dam is initially empty find W as a function of t .
(c) Find the volume of water in the dam at the end of the year.

(a) Let $R(t) = \frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$

$$\begin{aligned} R'(t) &= 0 - 2 \cos \frac{\pi}{12} t \sin \frac{\pi}{12} t \cdot \frac{\pi}{12} \\ R'(t) &= -\sin 2 \left(\frac{\pi}{12} t \right) \cdot \frac{\pi}{12} \\ R'(t) &= -\frac{\pi}{12} \sin \frac{\pi}{6} t \\ R'(t) &= 0 \quad 0 \leq t \leq 12 \\ -\frac{\pi}{12} \sin \frac{\pi}{6} t &= 0 \\ \sin \frac{\pi}{6} t &= 0 \\ \frac{\pi}{6} t &= 0, \pi, 2\pi \\ t &= 0, 6, 12 \\ t &= 0 \text{ or } 6 \text{ or } 12 \end{aligned}$$

$$\begin{aligned} t=0, R(0) &= 1.2 - \cos^2 \frac{\pi}{12}(0) = 1.2 - 1 = 0.2 \\ t=6, R(6) &= 1.2 - \cos^2 \frac{\pi}{12}(6) = 1.2 - 0 = 1.2 \\ t=12, R(12) &= 1.2 - \cos^2 \pi = 1.2 - 1 = 0.2 \end{aligned}$$

$\therefore$ Maximum flow rate is
 $R_{\max} = 1.2 \text{ thousand m}^3/\text{month}$

(b) $\frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$

$$\begin{aligned} \frac{dW}{dt} &= 1.2 - \left(\frac{1 + \cos \frac{\pi}{6} t}{2} \right) \\ &= 1.2 - 0.5 - \frac{1}{2} \cos \frac{\pi}{6} t \\ \frac{dW}{dt} &= 0.7 - 0.5 \cos \frac{\pi}{6} t \\ W &= \int \left(0.7 - 0.5 \cos \frac{\pi}{6} t \right) dt \\ W &= 0.7t - 0.5 \frac{\sin \frac{\pi}{6} t}{\frac{\pi}{6}} + C \\ W &= 0.7t - \frac{3}{\pi} \sin \frac{\pi}{6} t + C \\ t=0, W &= 0 \\ 0 &= 0 - 0 + C \Rightarrow C = 0 \\ \therefore W(t) &= 0.7t - \frac{3}{\pi} \sin \frac{\pi}{6} t \end{aligned}$$

② end of the year, $t=12$

$$w(12) = 0.7(12) - \frac{3}{\pi} \sin \frac{\pi}{6}(12)$$

$$= 8.4 \text{ thousand } m^3$$

$$w(12) = 8400 m^3$$

Exponential Growth and Decay

46. A radioactive substance decomposes from 100g to 90g in 2 hours according to the formula $M = M_0 e^{-kt}$, where t is the number of hours. Find:

- (a) k correct to 3 decimal places,
 (b) the half life of the substance (ie. the time taken for it to decay to half its original mass). Give your answer to the nearest hour.

(a) $M = 100 e^{-kt}$
 $t=2, M(2) = 90$
 $100 e^{-2k} = 90$
 $e^{-2k} = 0.9$
 $-2k = \ln(0.9)$
 $k = \frac{\ln(0.9)}{-2}$
 $k \approx 0.053/h$

(b) $M = 50$
 $100 e^{-kt} = 50$
 $e^{-kt} = 0.5$
 $-kt = \ln(0.5)$
 $t = \frac{\ln(0.5)}{-k}$
 $t = \frac{\ln(0.5)}{-\left(\frac{\ln(0.9)}{-2}\right)}$
 $\therefore t = \frac{2 \ln(0.5)}{\ln(0.9)}$
 $t \approx 13.158$
 $t \approx 13 \text{ hours}$
 (Half life)

47. The rate of change of bacteria in a container is given by $\frac{dN}{dt} = kN$, where N is the population of bacteria in the container and t is the time in hours.

(a) Show that $N = N_0 e^{kt}$, where N_0 is a constant, is a solution to the differential equation $\frac{dN}{dt} = kN$.

(b) If initially there are 2000 bacteria in the container and the bacteria increase at the rate of 2% of the current population, find the number of bacteria in the container after 100 hours.

(a) $N = N_0 e^{kt}$
 $\frac{dN}{dt} = N_0 \cdot e^{kt} \cdot k = kN$
 $\frac{dN}{dt} = kN$

(b) $N_0 = 2000$
 $k = 0.02/h$
 $\therefore N = 2000 e^{0.02t}$
 $t=100, N(100) = 2000 e^{0.02(100)}$
 $= 14778 \text{ bacteria}$

Exponential Growth and Decay continued

48. The number of bacteria N in a colony after t minutes is given by $N = 10\,000 e^{0.05t}$.
Find:

- (i) the number of bacteria after 10 minutes.
- (ii) the time required for the original number to double.
- (iii) the rate at which the colony increases when:
 - (a) $t = 10$.
 - (b) $N = 20\,000$.

(i) $N(t) = 10\,000 e^{0.05t}$
 $t = 10, N(10) = 10\,000 e^{0.05(10)} = 10\,000 e^{0.5} \approx 16\,487 \text{ bacteria}$

(ii) $N = 20\,000$
 $10\,000 e^{0.05t} = 20\,000 \Rightarrow e^{0.05t} = 2 \Rightarrow 0.05t = \ln 2$
 $t = \frac{\ln 2}{0.05} \approx 13.863 \text{ min}$

(iii) $N(t) = 10\,000 e^{0.05t}$
 $N'(t) = 10\,000 e^{0.05t} \cdot 0.05 = 0.05N$
 $N'(t) = 500 e^{0.05t}$
 (a) $t = 10, N'(10) = 500 e^{0.05(10)} \approx 824.361 \text{ bacteria/min}$

(b) $N'(t) = 0.05N$
 when $N = 20\,000, N' = 0.05(20\,000)$
 $= 1000 \text{ bacteria/min}$

Exponential Growth and Decay continued

49. A heated body is cooling and the excess $\theta^\circ\text{C}$ of its temperature above that of its surroundings is given by $\theta = Ae^{-kt}$ where t is the time in minutes.

(i) If at time $t=0$, $\theta=80$, find A .

(ii) If the temperature of the surroundings is 20°C and the body cools to 70°C in 10 minutes, find

(a) its temperature after 20 minutes.

(b) the time taken to reach 60°C .

$$(i) \theta(t) = Ae^{-kt}$$

$$t=0, \theta=80$$

$$80 = A \cdot e^{-k(0)} \Rightarrow A = 80$$

$$\therefore \theta(t) = 80e^{-kt}$$

$$(ii) T = 20 + \theta$$

$$T = 20 + 80e^{-kt}$$

$$\text{given } T(10) = 70$$

$$70 = 20 + 80e^{-10k}$$

$$50 = 80e^{-10k}$$

$$e^{-10k} = \frac{5}{8}$$

$$\Rightarrow -10k = \ln\left(\frac{5}{8}\right)$$

$$k = \frac{-1}{10} \ln\left(\frac{5}{8}\right)$$

$$(ii)$$

$$(b) T = 60$$

$$20 + 80e^{-kt} = 60$$

$$80e^{-kt} = 40$$

$$\Rightarrow e^{-kt} = \frac{1}{2}$$

$$-kt = \ln\left(\frac{1}{2}\right)$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-\left(\frac{-1}{10} \ln\left(\frac{5}{8}\right)\right)}$$

$$t = \frac{10 \ln\left(\frac{1}{2}\right)}{\ln\left(\frac{5}{8}\right)}$$

$$t \approx 14.75 \text{ minutes}$$

$$(a) \text{ After 20 minutes } t=20$$

$$T(t) = 20 + 80e^{-kt}$$

$$T(20) = 20 + 80e^{-\left(\frac{-1}{10} \ln\left(\frac{5}{8}\right)\right) \cdot 20}$$

$$= 20 + 80 \cdot \frac{25}{64}$$

$$= 51.25^\circ\text{C}$$

Implicit Differentiation

50. Find $\frac{dy}{dx}$ in each of the following:

(g) $x^3 y^2 = 3$

(h) $xy + y^2 + 6 = 0$

(i) $x^2 y^3 + xy^2 - 8 = 0$

(j) $x^4 + 5x^2 y - 6y^2 x^3 + 3x = 0$

(k) $\frac{x}{y} + 4x = 0$

(l) $x^n y^m = mn$

(g) $x^3 \cdot 2y \cdot y' + y^2 \cdot 3x^2 = 0$
 $y' = \frac{-3x^2 y^2}{2x^3 y} = \frac{-3y}{2x}$

(j) $4x^3 + 5x^2 y' + y \cdot 10x - 6y^2 3x^2 - x^3 12y y' + 3 = 0$
 $y'(5x^2 - 12x^3 y) = -3 - 4x^3 - 10xy + 18x^2 y^2$
 $y' = \frac{18x^2 y^2 - 4x^3 - 10xy - 3}{5x^2 - 12x^3 y}$

(h) $x \cdot y' + y \cdot 1 + 2y y' + 0 = 0$
 $y'(x + 2y) = -y$
 $y' = \frac{-y}{x + 2y}$

(k) $\frac{y \cdot 1 - x y'}{y^2} + 4 = 0$
 $y - x y' = -4y^2$
 $\frac{y + 4y^2}{x} = y'$

(i) $x^2 \cdot 3y^2 y' + y^3 \cdot 2x + x \cdot 2y y' + y^2 \cdot 1 = 0$
 $y'(3x^2 y^2 + 2xy) = -y^2 - 2xy^3$
 $y' = \frac{-y^2(1 + 2x)}{xy(3xy + 2)} = \frac{-y(1 + 2x)}{x(3xy + 2)}$

(l) $x^n \cdot m y^{m-1} y' + y^m \cdot n x^{n-1} = 0$
 $y' = \frac{-n \cdot y^m \cdot x^{n-1}}{m \cdot x^n \cdot y^{m-1}}$

51. If $x = y^2 + y$ find $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$.

$y' = \frac{-n}{m} \cdot \frac{y}{x} \Rightarrow y' = \frac{-n y}{m x}$

$x = y^2 + y$

$1 = 2y \cdot \frac{dy}{dx} + \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{1}{2y+1}$

$\frac{dy}{dx} = (2y+1)^{-1}$

$\frac{d^2 y}{dx^2} = (-1)(2y+1)^{-2} \left(2 \cdot \frac{dy}{dx}\right)$
 $= \frac{-2}{(2y+1)^2} \cdot \frac{1}{(2y+1)}$

$\frac{d^2 y}{dx^2} = \frac{-2}{(2y+1)^3}$

Implicit Differentiation continued

52. For the curve $(x+y)^2 = 4x^2y + 1$ find the equation of the tangent at the point $(1,2)$ and the equation of the normal at the point $(1,0)$.

$$\frac{d}{dx} [(x+y)^2] = \frac{d}{dx} [4x^2y + 1]$$

$$2(x+y) \cdot \left(1 + \frac{dy}{dx}\right) = 4x^2 \cdot \frac{dy}{dx} + y \cdot 8x + 0$$

$$(x+y) + (x+y) \frac{dy}{dx} = 2x^2 \frac{dy}{dx} + 4xy$$

$$\frac{dy}{dx} (x+y-2x^2) = 4xy - x - y$$

$$\frac{dy}{dx} = \frac{4xy - x - y}{x + y - 2x^2}$$

$$\text{at } (1,2), m_T = \frac{8 - 1 - 2}{1 + 2 - 2} = \frac{5}{1} = 5$$

$$\text{at } (1,0), m_N = \frac{-1}{m_T} = \frac{-1}{\frac{5-1}{1-2}} = -1$$

53. Find $\frac{dy}{dx}$ in each of the following:

(a) $e^x + e^y = e^5$

$$\frac{d}{dx} [e^x + e^y] = \frac{d}{dx} (e^5)$$

$$e^x + e^y \frac{dy}{dx} = 0$$

$$e^y \frac{dy}{dx} = -e^x$$

$$\frac{dy}{dx} = \frac{-e^x}{e^y}$$

$$\frac{dy}{dx} = -e^{x-y}$$

(b) $xe^y = y^2 + e^x$

$$\frac{d}{dx} (xe^y) = \frac{d}{dx} (y^2 + e^x)$$

$$x \cdot e^y \frac{dy}{dx} + e^y \cdot 1 = 2y \frac{dy}{dx} + e^x$$

$$\frac{dy}{dx} (xe^y - 2y) = e^x - e^y$$

$$\frac{dy}{dx} = \frac{e^x - e^y}{xe^y - 2y}$$

Equation of the tangent at $(1,2)$:

$$y - 2 = 5(x - 1)$$

$$y = 5x - 3$$

Equation of Normal at $(1,0)$:

$$y - 0 = -1(x - 1)$$

$$y = -x + 1$$

$$y = 1 - x$$

Implicit Differentiation continued

54. Find $\frac{dy}{dx}$ in each of the following :

(a) $x \ln y + y = 20$

$$\frac{d}{dx} [x \ln y + y] = \frac{d}{dx} (20)$$

$$x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \ln y \cdot 1 + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} \left(\frac{x}{y} + 1 \right) = -\ln y$$

$$\frac{dy}{dx} \left(\frac{x+y}{y} \right) = -\ln y$$

$$\frac{dy}{dx} = \frac{-y \ln y}{x+y}$$

(b) $\ln x \ln y = 1$

$$\frac{d}{dx} (\ln x \cdot \ln y) = \frac{d}{dx} (1)$$

$$\ln x \cdot \frac{1}{y} \frac{dy}{dx} + \ln y \cdot \frac{1}{x} = 0$$

$$\frac{dy}{dx} = \frac{-\frac{\ln y}{x}}{\frac{\ln x}{y}}$$

$$\frac{dy}{dx} = \frac{-y \ln y}{x \ln x}$$

55. Find $\frac{dy}{dx}$ in each of the following :

(c) $\sin(x-2y) = y$

$$\frac{d}{dx} [\sin(x-2y)] = \frac{d}{dx} (y)$$

$$\cos(x-2y) \cdot \left(1 - 2 \frac{dy}{dx} \right) = \frac{dy}{dx}$$

$$\cos(x-2y) - 2 \cos(x-2y) \frac{dy}{dx} = \frac{dy}{dx}$$

$$\frac{dy}{dx} (1 + 2 \cos(x-2y)) = \cos(x-2y)$$

$$\therefore \frac{dy}{dx} = \frac{\cos(x-2y)}{1 + 2 \cos(x-2y)}$$

(d) $\sin(y+x) = \cos(y-x)$

$$\frac{d}{dx} [\sin(y+x)] = \frac{d}{dx} [\cos(y-x)]$$

$$\cos(y+x) \cdot \left(\frac{dy}{dx} + 1 \right) = -\sin(y-x) \left[\frac{dy}{dx} - 1 \right]$$

$$\frac{dy}{dx} [\cos(y+x) + \sin(y-x)] = \sin(y-x) - \cos(y+x)$$

$$\frac{dy}{dx} = \frac{\sin(y-x) - \cos(y+x)}{\cos(y+x) + \sin(y-x)}$$

Related Rates

56. A spherical balloon is being filled with air at the rate of $100 \text{ cm}^3/\text{min}$. Find the rate at which the radius of the sphere is increasing when the radius is 5 cm .

Given $\frac{dV}{dt} = 100 \text{ cm}^3/\text{min}$

Find $\frac{dr}{dt}$ when $r = 5 \text{ cm}$

Volume of a sphere:

$$V = \frac{4\pi}{3} r^3$$

$$\frac{dV}{dt} = \frac{4\pi}{3} \cdot 3r^2 \cdot \frac{dr}{dt}$$

$$100 = 4\pi \cdot (5)^2 \cdot \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ cm/min}$$

$$\therefore \frac{dr}{dt} = \frac{1}{\pi} \text{ cm/min}$$

57. A ladder 5 metres long has its upper end against a vertical wall and its lower end on the horizontal floor. The lower end moves away from the wall at a constant rate of 1 m/second . Find the rate at which the upper end moves down the wall when the lower end is 4 metres from the wall.

$$x^2 + y^2 = 25$$

$$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(25)$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

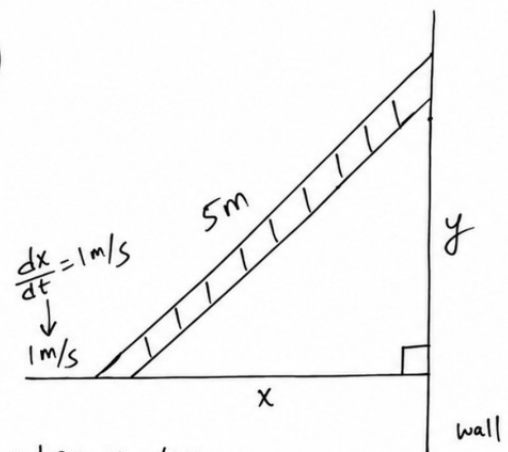
$$y \cdot \frac{dy}{dt} = -x \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$$

$$= -\frac{4}{3} \cdot (1)$$

$$\frac{dy}{dt} = -\frac{4}{3} \text{ m/s}$$

$\therefore$ upper end moves down at $\frac{4}{3} \text{ m/s}$



when $x = 4 \text{ m}$

$$5^2 = x^2 + y^2$$

$$25 - 16 = y^2$$

$$y = 3$$

Related Rates continued

58. Sand falling forms a conical pile such that the height of the pile is one and a half times the radius.

- (a) Show that the volume of the pile is $V = \frac{4}{27} \pi h^3$.
- (b) If the sand is falling at the rate of $\frac{\pi}{10} \text{ m}^3 / \text{minute}$, find the rate at which the height is increasing when the pile of sand is 3m high.

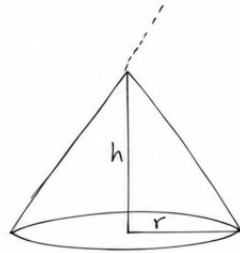
(a)

$$h = 1\frac{1}{2}r$$

$$h = \frac{3r}{2} \Rightarrow r = \frac{2h}{3}$$

Volume of a cone,

$$\begin{aligned} V &= \frac{h}{3} \cdot \pi r^2 \\ &= \frac{h}{3} \pi \left(\frac{2h}{3}\right)^2 \\ &= \frac{h}{3} \cdot \pi \cdot \frac{4h^2}{9} \\ &= \frac{4\pi h^3}{27} \\ V &= \frac{4\pi h^3}{27} \end{aligned}$$



(b)

$$\text{find } \frac{dh}{dt} \text{ when } h = 3\text{m}$$

$$\text{given } \frac{dV}{dt} = \frac{\pi}{10} \text{ m}^3/\text{min}$$

$$V = \frac{4}{27} \pi h^3$$

$$\frac{d}{dt}(V) = \frac{d}{dt}\left(\frac{4\pi}{27} h^3\right)$$

$$\frac{dV}{dt} = \frac{4\pi}{27} \cdot 3h^2 \cdot \frac{dh}{dt}$$

$$\frac{\pi}{10} = \frac{4\pi}{9} \cdot (3)^2 \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{1}{40} \text{ m/min}$$

59. A searchlight on the ground detects and tracks a plane that is due east of the searchlight and at a height of 900 m above the ground. The plane is travelling due West at 240 km per hour. Find the rate of change of the angle of elevation of the plane from the searchlight when the plane is 900 m east of the searchlight.

$$\tan \theta = \frac{900}{x}$$

$$\frac{dx}{dt} = -240 \text{ km/hr}$$

$$= \frac{-240 \times 1000}{3600}$$

$$= -\frac{200}{3} \text{ m/s}$$

Angle increases

as the plane approaches

$$\frac{d}{dt}(\tan \theta) = \frac{d}{dt}\left(\frac{900}{x}\right)$$

$$\sec^2 \theta \cdot \frac{d\theta}{dt} = 900 \left(-\frac{1}{x^2}\right) \cdot \frac{dx}{dt}$$

$$\text{When } x=900, \theta = \frac{\pi}{4}$$

$$\text{if } \frac{d\theta}{dt} = -\frac{900}{900^2} \cdot \frac{200}{3}$$

$$= -\frac{1}{900} \cdot \frac{200}{3}$$

$$= -\frac{1}{27} \text{ rad/sec}$$

